ParFlow System Basics

How to setup and run a model and where to get help

ParFlow Short Course

Module 2

Learning Objectives:

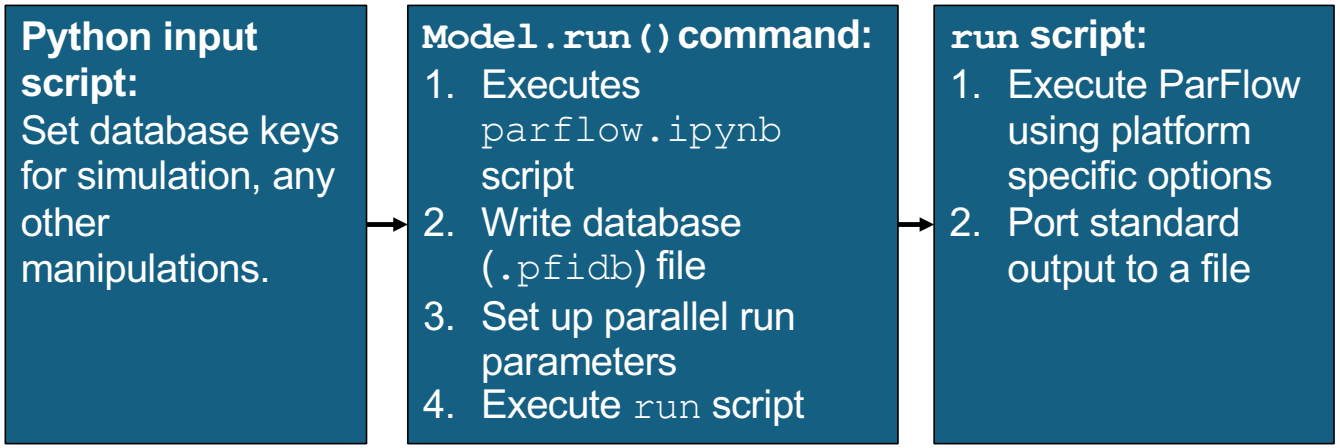
At the end of this module students will understand:

- That there are a suite of options for working with ParFlow models
- ParFlow models can be setup and run using TCL or Python (Or directly from a PFIDB)
- The difference between setting keys and running a model
- Options for installing and running ParFlow on multiple systems
- What PFTools are and how you can download and access them
- What the PFTools python package is and how to access it
- Where to get help on ParFlow Keys and PFTools
- Where to find example parflow scripts in tcl and python

Module Outline:

1. *Setting up a model input file*
2. *Running a model*
3. *What comes out and how to look at it*
4. *Tools for working with ParFlow models*
5. *Online resources*

1. Setting up a model input file *(How you tell ParFlow what to do)*



Input Scripts

- TCL/TK scripting language with python interface
- All parameters input as keys using `pfset` command (tcl) or `model.”key_name”` (python)
- Keys used to build a database that ParFlow uses
- ParFlow executed by `pfrun` command (for tcl) and `model.run()` for python
- Since input file is a script may be run like a program

Example: Setting up the input grid

```
#---------------------------------------------------------
# Computational Grid
#---------------------------------------------------------
#Locate the origin in the domain.
model.ComputationalGrid.Lower.X = 0.0
model.ComputationalGrid.Lower.Y = 0.0
model.ComputationalGrid.Lower.Z = 0.0

# Define the size of each grid cell. The length units are
the same as those on hydraulic conductivity, here that is
meters.
model.ComputationalGrid.DX = 1000.0
model.ComputationalGrid.DY = 1000.0
model.ComputationalGrid.DZ = 200.0

# Define the number of grid blocks in the domain.
model.ComputationalGrid.NX = 64
model.ComputationalGrid.NY = 32
model.ComputationalGrid.NZ = 10
```

Coordinates (length units)

Cell size (length units)

Grid dimensions (integer)

Example: Setting up the timing

```
#---------------------------------------------------------
# Setup timing info
#---------------------------------------------------------

model.TimingInfo.BaseUnit = 1.0

model.TimingInfo.StartCount = 0

model.TimingInfo.StartTime = 0
model.TimingInfo.StopTime = 72.0

model.TimingInfo.DumpInterval = 1.0

model.TimeStep.Type = "Constant"

model.TimeStep.Value = 1.0
```

Sets time units for time cycles (T)

Initial output file number

Start and finish time for simulation (T)

Interval to write output (T)
-1 outputs at every timestep

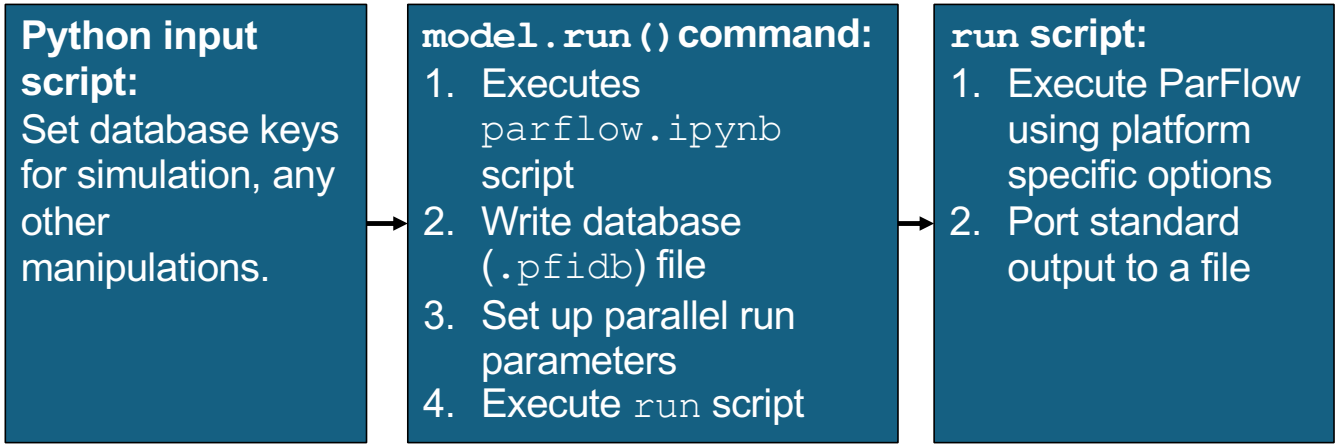
Timestep type

ΔT (T)

Best practices for building an input file:

- Start from an existing script:
  - Look at the annotated input scripts in the manual
  - Look at the test problems that come with ParFlow (See list in section 3.5)
  - Use scripts for test problems presented in this course
- Get the details on every input key from the manual (Section 6)

2. Running ParFlow simulations (*How to press go*)



Running ParFlow

- `model.run()` command
  - Builds database of keys
  - Executes program
- Some error checking of keys
- Actual command line runs executable or mpirun's executable
- May run parflow code more than once in single script
- Need parflow package/header information

Running ParFlow (input file)

Mandatory Content at the top of your python script:

```
# Import required packages

import os
import numpy as np
from parflow import Run
import shutil from parflow.tools.fs
import mkdir, cp, get_absolute_path, exists from
parflow.tools.settings
import set_working_directory
…
model.run()
```

Load the recommended packages to run parflow in python

Run parflow, execute run command on model object

To run the model:

model.run()

3. Handling outputs
(*What comes out and how to look at it*)

Running ParFlow (file structure)

- Project name is the base for all output
- Most output is *project*.out.*var*.*time*.ext

For a project called ‘myrun’

Log files:

```
myrun.out.log
myrun.out.kinsol.log
```

Pressure/Saturation files:

```
myrun.out.press.00001.pfb
myrun.out.satur.00001.pfb
```

Output time step, 00000 is initial, integer values depending on output times

Mask file:

```
myrun.out.mask.00000.pfb
```

The mask is a file of zero’s and ones, 0=inactive cell, 1=active cell

Perm/porosity files:

```
myrun.out.perm_x.pfb
myrun.out.porosity.pfb
```

Other/diagnostic files:

```
myrun.pfidb
myrun.out.pftcl
myrun.out.txt
```

Parflow database

Line output

Visualizing Outputs

Needs to be updated with Python stuff

ParaView:

- Free, developed by Kitware Inc
- https://paraview.org
- VTK and pfb formats supported

Visit:

- Free, developed at LLNL
- (http://www.llnl.gov/visit/)
- VTK and SILO format, which has many options within ParFlow (converting or IO), fully-supported

Need to make this section

4. Tools for working with ParFlow models

PFTools

- TCL keys that you can use to manipulate ParFLow inputs and outputs:
  - Extract parts of your domain to look at
  - Calculate water balance components (hydrology module)
  - Convert pfb outputs to other file types

5. Online resources

Need to make this section